Hoonki Yeo

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RESEARCH INTERESTS

Optimization under uncertainty; optimization for machine learning; optimization for operational decision-making.

EDUCATION

Georgia Institute of Technology Aug 2025 - Dec 2026 (expected)
M.S. in Operations Research, H. Milton Stewart School of Industrial and Systems Engineering Atlanta, GA

- **Selected coursework:** Convex Optimization, Linear Optimization, Simulation, Theoretical Statistics, High-Dimensional Data Analytics, Computational Data Analytics.

University of Wisconsin-Madison Aug 2019 - Aug 2022
B.S. in Data Science Madison, WI

- **GPA:** 3.84/4.00. Graduated with Distinction; Dean's List for four semesters.

RESEARCH EXPERIENCE

Derivative-Free Stochastic Optimization for Constrained Learning Mar 2026 - Present
Georgia Institute of Technology; Advisor: Prof. Sen Na Atlanta, GA

- Formulated a partially linear single-index learning problem as equality-constrained stochastic optimization, with noisy objective evaluations arising from neural-network training.
- Implemented derivative-free stochastic sequential quadratic programming (DF-SSQP), verified the implementation numerically, and examined the effects of constraint-Jacobian estimation on optimization progress and feasibility.
- Designing and conducting controlled experiments to assess the value of optimizing the index direction relative to fixed-direction models, and to characterize DF-SSQP’s behavior, strengths, and limitations through comparisons with zeroth- and first-order methods.

Optimizing 20-Minute Transit Accessibility under Uncertainty Sep 2026 - Present
Data4Good Analytics Competition · Team Research Project

- Compiled and mapped bus, light rail, and streetcar networks in Tempe, Arizona, using GTFS data and checking stop coverage against the municipal boundary.
- Designing a residential-unit-weighted analysis of 20-minute access to daily destinations, accounting for walking, waiting, in-vehicle travel, and transfers.
- Designing an optimization study of targeted transit service adjustments to improve accessibility under uncertain travel times and transfer conditions.

PROFESSIONAL EXPERIENCE

Data Scientist
2024 - 2025

Aivelabs
Seoul, South Korea

- Developed and deployed production machine learning workflows for customer value forecasting, from data preparation and model training to scoring.
- Deployed personalized recommendations and a multi-armed bandit feedback loop for outreach targeting.
- Developed customer segmentation and KPI dashboards to support marketing and sales decisions in collaboration with cross-functional teams.

Data Analyst
2022 - 2023

Retentics
Seoul, South Korea

- Developed an internal Python preprocessing package to standardize data preparation workflows.
- Maintained data-processing and validation pipelines and conducted analyses for client projects to support operational decisions.

TEACHING EXPERIENCE

Peer Tutor
2020 - 2021

University of Wisconsin-Madison
Madison, WI

- Provided one-on-one tutoring in introductory statistics and probability for STAT 301 and STAT 309.

LEADERSHIP & SERVICE

Sergeant, Thermal Observation Device Operator
2016 - 2018

Republic of Korea Army
Yeoncheon, South Korea

- Led a small team responsible for observation-equipment readiness and training new personnel.

TECHNICAL SKILLS

Programming: Python, R, SQL.

Optimization: Gurobi, OR-Tools.

Scientific computing and modeling: NumPy, PyTorch, scikit-learn, TensorFlow.

Data systems: Apache Airflow, Apache Spark, MySQL, PostgreSQL, Tableau, AWS.

Research tools: Git, LaTeX.